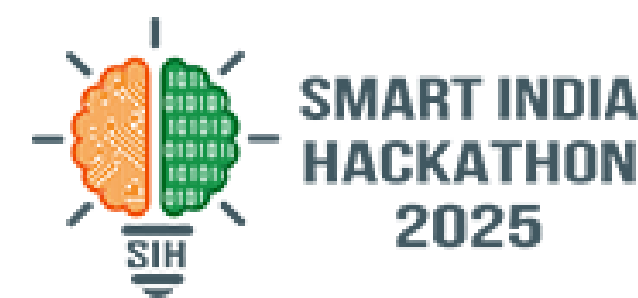




SMART INDIA HACKATHON 2025



PLUNO

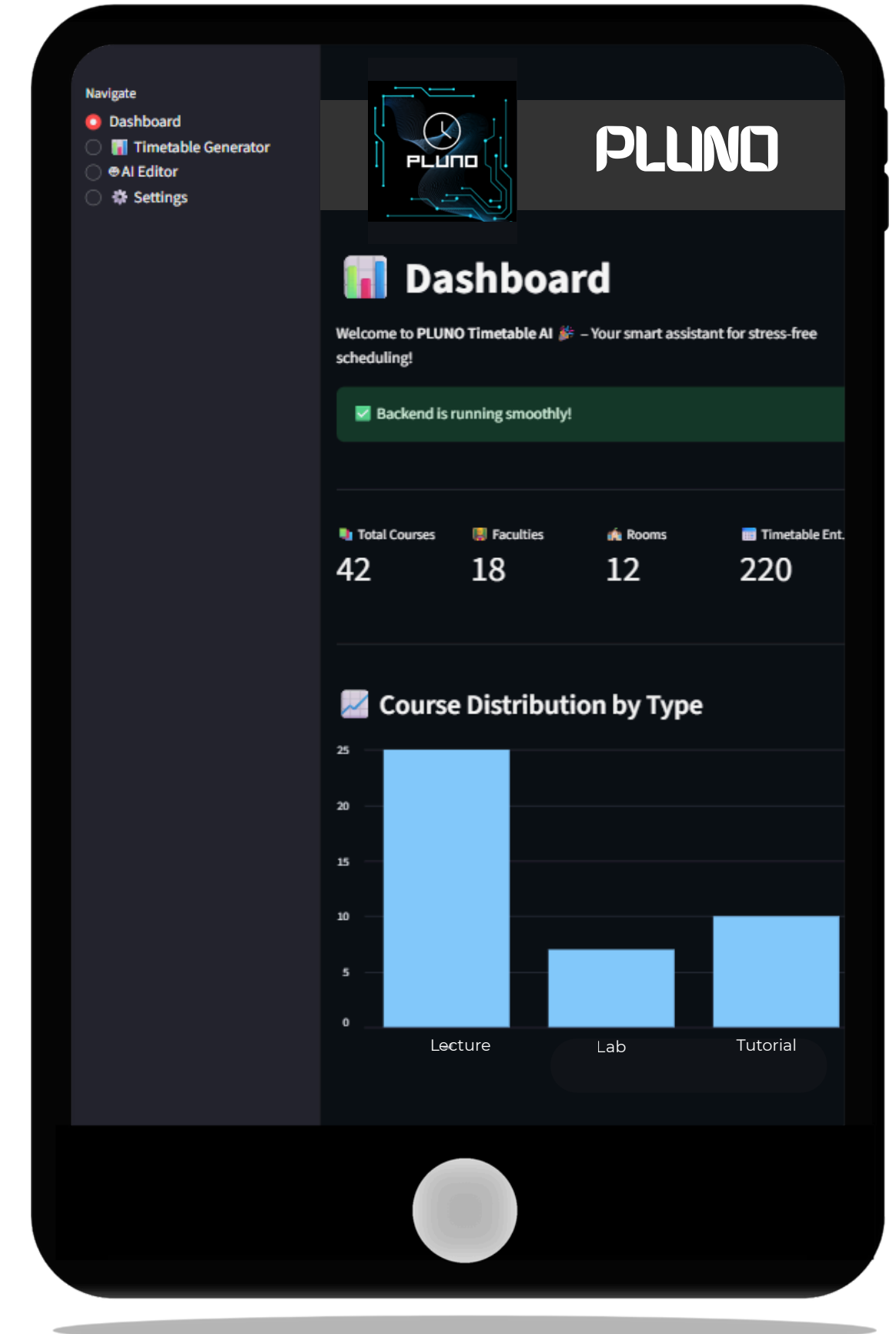
YOUR ONE STOP FOR SMART PLANNING

- **Problem Statement ID** – 25091
- **Problem Statement Title** - AI-Based Timetable Generation System aligned with NEP 2020 for Multidisciplinary Education Structures
- **Theme** - Smart Automation
- **PS Category** - Software
- **Team Name** - Unify



PROPOSED SOLUTION

- An intelligent timetable generation system aligned with **NEP 2020** for optimized academic scheduling.
- Integrates key factors:
 - Student course selection based on preferences.
 - Faculty availability, including substitution management.
 - Room and lab resource constraints.
- Supports dynamic editing and real-time updates through a user-friendly admin interface.
- Offers scenario simulation to test and optimize scheduling options.
- Includes an **AI chatbot** to assist users and streamline interactions.
- Designed for scalability, ensuring efficient resource utilization and flexible schedule management.



FEASIBILITY

- **TECHNICAL**

Proven **optimization algorithms and existing data** (students, faculty, rooms) make system development realistic.

- **POLICY**

Fully aligned with **NEP 2020** emphasis on digital tools and flexible credit-based structures.

- **OPERATIONAL**

Easy data imports and a **user-friendly interface** ensure adoption by timetable committees with minimal training.

- **INTEGRATION**

Can connect with existing Academic Management Systems (via API/CSV) and scale to large institutions.

VIABILITY

- **ECONOMIC**

Saves significant faculty/admin hours, reduces clashes, and offers strong ROI within 1–2 years.

- **SCALABILITY**

Adaptable for future NEP updates, new courses, and multi-campus institutions.

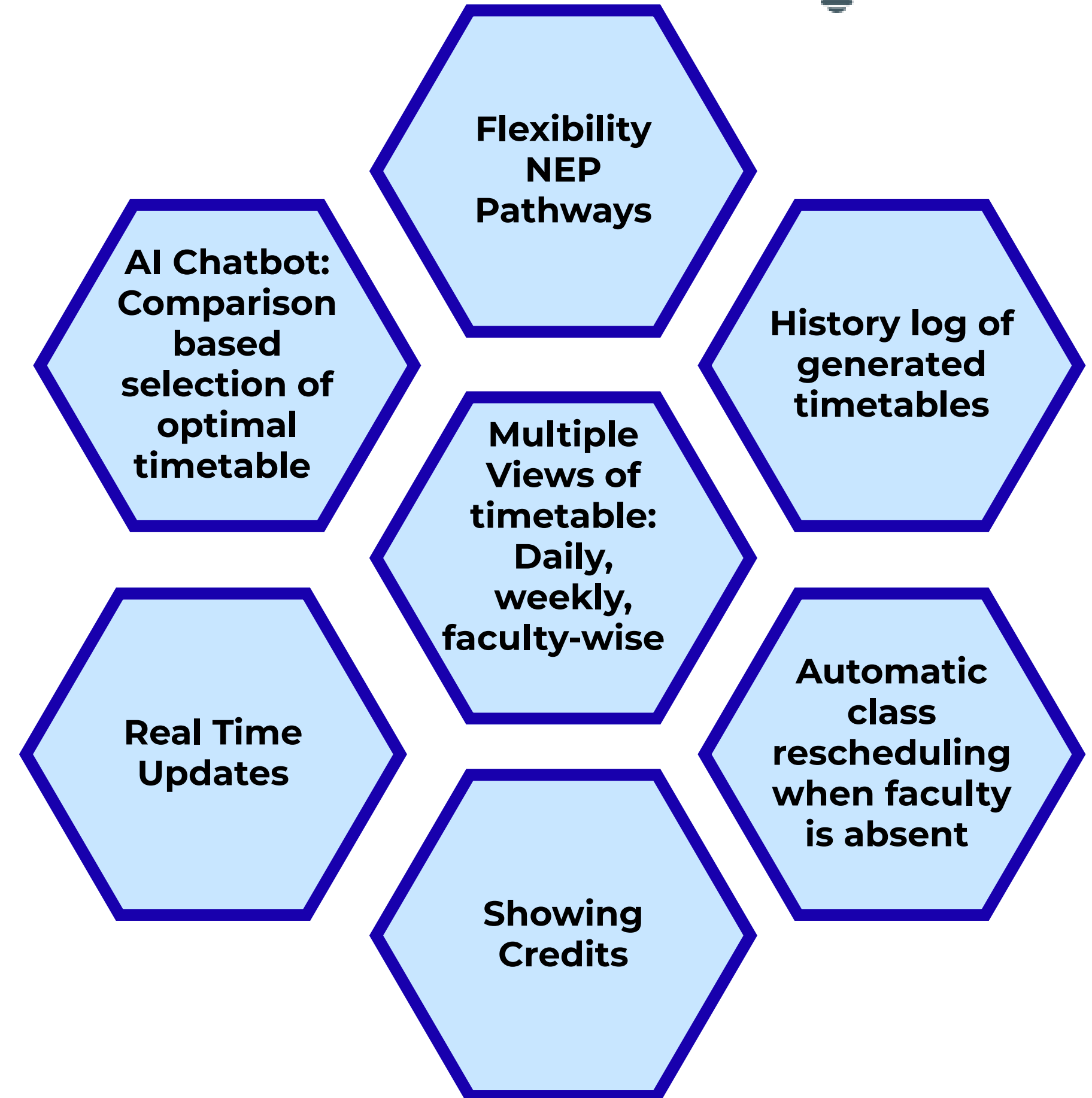
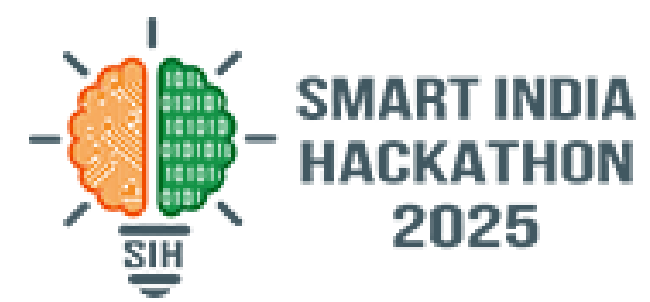
- **ACADEMIC**

Ensures smooth implementation of B.Ed., M.Ed., ITEP, and elective flexibility with minimal conflicts.

- **SUSTAINABILITY**

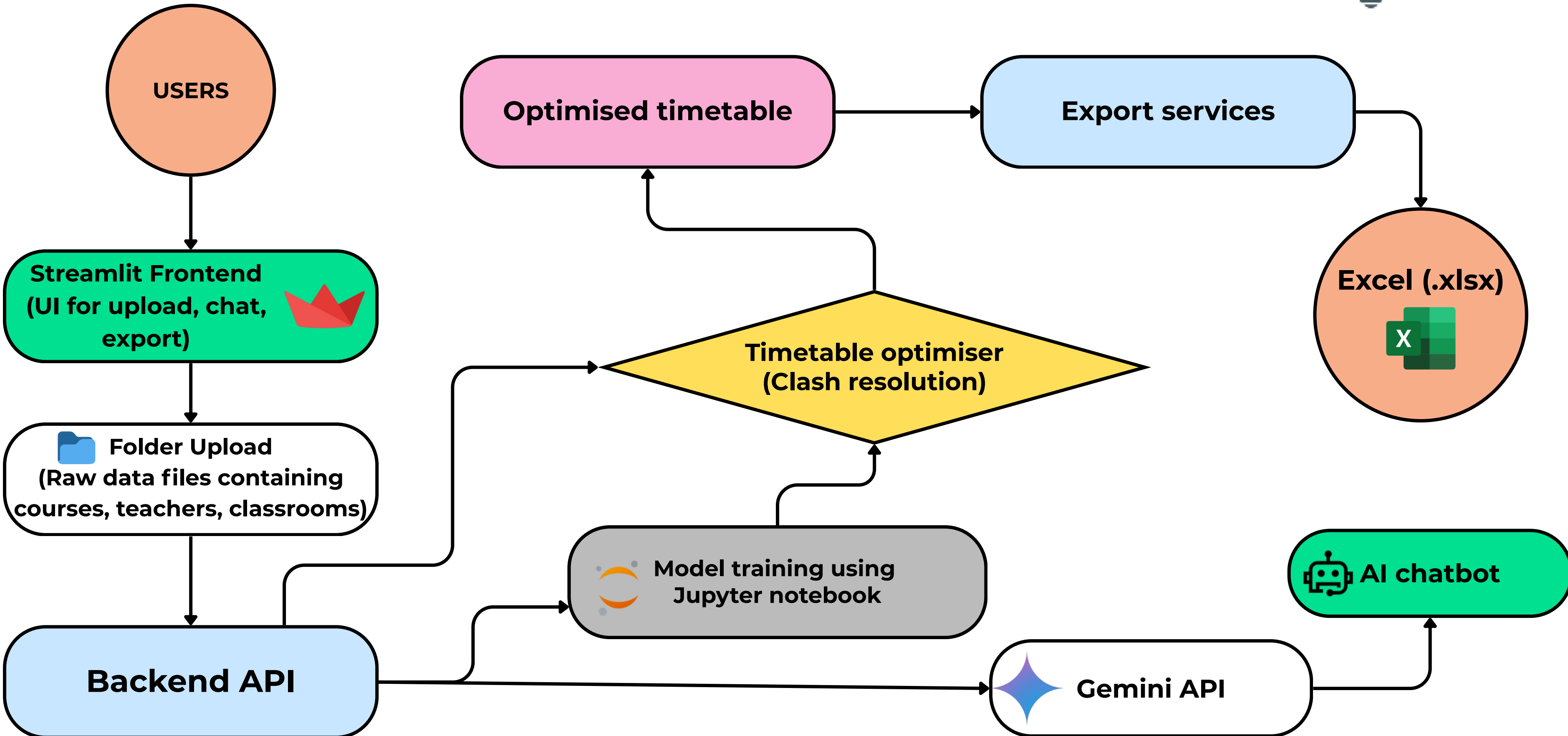
Ensure long-term usability and institutional acceptance.

FEATURES



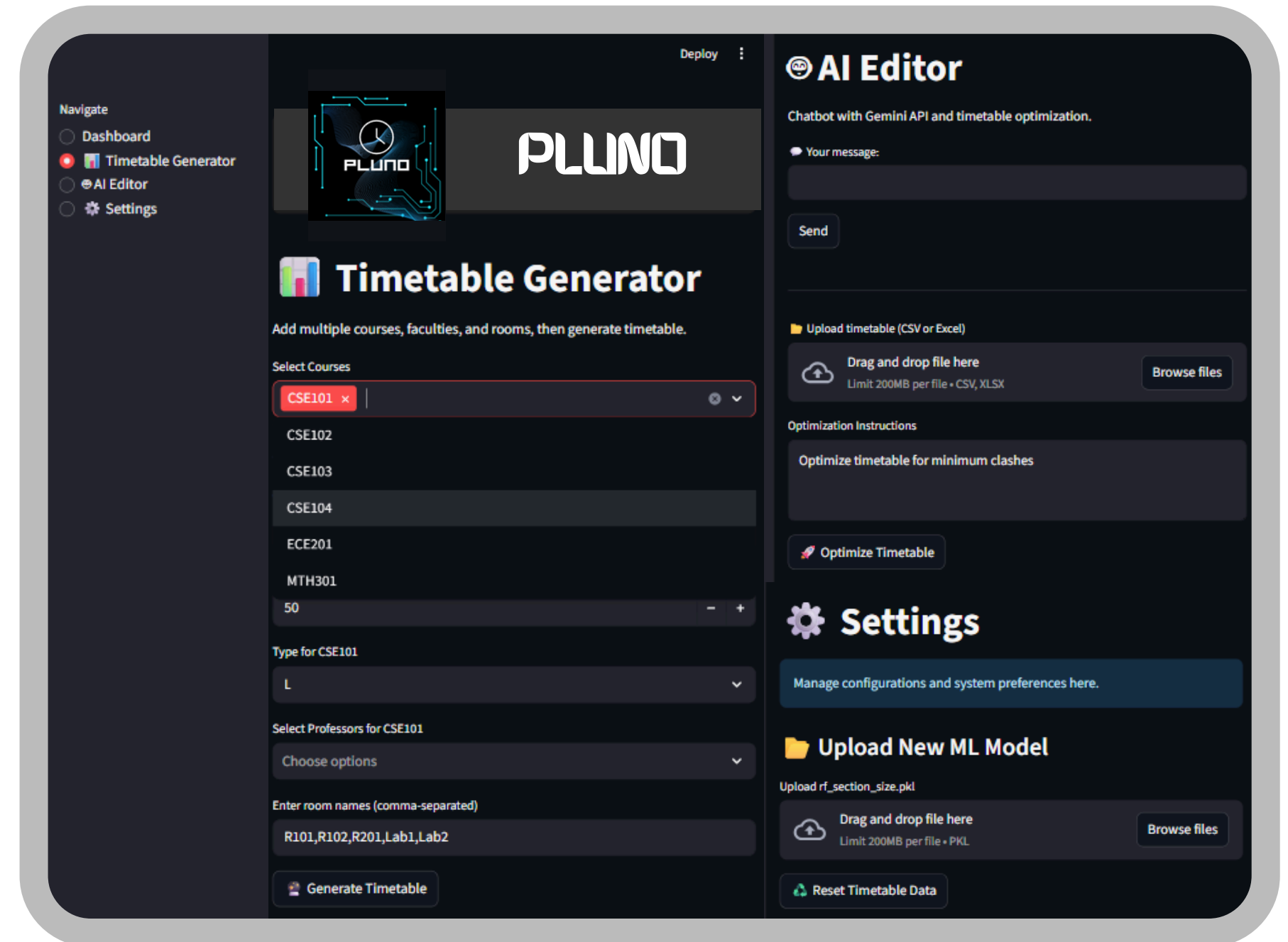
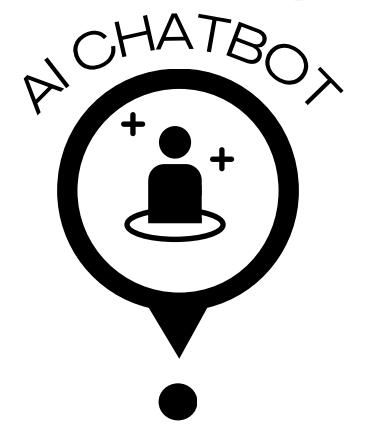
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TECHNICAL APPROACH



IMPACT AND BENEFITS

- This AI-based timetable planner is meant for colleges, university and school administration, to assist them in building optimised timetables
- Keep in mind all factors such as teacher and room availability, the courses chosen by the student, credit structure etc. in alignment to NEP 2020.
- Our solution is a technology based idea that aims to solve the intensive task of creating timetables for dozens of classes.
- This application is easy to understand and run, thus reducing the amount of manpower ordinarily required for this task.



Challenges

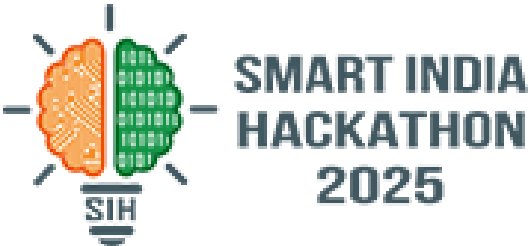
Strategies

1.	Data Fragmentation & Inconsistency – Student electives, faculty workload, and curriculum records exist in scattered formats (manual entries, Excel, different AMS), leading to errors & delays.	Build a centralized digital database with standardized formats; enable API integration with existing Academic Management Systems ; use secure cloud platforms for seamless access.
2.	High Complexity of Scheduling - Managing electives, credits, ratios, internships, and faculty availability makes manual timetabling highly intensive .	Apply AI/ML-driven optimization algorithms (e.g., genetic algorithms, constraint programming) to handle constraints; use scenario simulation tools to test multiple timetable versions.
3.	Infrastructure & Resource Limitations – Few classrooms/labs and uneven faculty distribution cause clashes , worsened by sudden absences or room issues	Incorporate real-time editing and auto-rescheduling features; build conflict detection alerts ; allow flexible last-minute adjustments without disrupting the entire timetable.
4.	Scalability & Policy Evolution – NEP 2020’s flexibility and future changes in courses, credits, and pathways demand frequent updates .	Use a modular, scalable system architecture; allow plug-and-play course components; maintain a continuous feedback mechanism with institutions to adapt to policy changes quickly.

RESEARCH & REFERENCES

- https://www.researchgate.net/publication/326281264_Unleashing_the_Potential_of_Chatbots_in_Education_A_State-Of-The-Art_Analysis
- https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_Final_English.pdf
- https://www.researchgate.net/publication/387463115_Optimizing_Course_Scheduling_with_Genetic_Algorithms_A_Dynamic_Approach
- Data in form of CSV files

OUR TEAM



BRANCH : COSE

1. SHRESHTH VERMA	ML and Backend
2. NIPUN MAHAJAN	Backend and Design
3. ISHWINDER KAUR	Frontend
4. ARADHYA GOYAL	Presentation
5. PRANAV TIWARI	Frontend
6. ARSHPREET KAUR	Design & Presentation